Critical Path Analysis – Practice

Grade 9 EMS. Four questions, easiest first. Answer key at the back – don't peek until you've tried!

Remember

- Node = circle: number on the left, EST top right, LFT bottom right.
- Letter above the line, duration below. Lines with the same prerequisites meet in one node.
- Forward pass (EST): add durations, take the **highest** into a node.
- Backward pass (LFT): subtract durations, take the **lowest** out of a node.
- Critical path = nodes where EST = LFT. Mark with //.
- Float = LFT (end node) − EST (start node) − Duration.

CPA practice Name: _____

Question 1: Warm-up

A short project with five activities. Use it to practise the drawing steps.

Activity	Description	Prerequisite(s)	Duration (days)
A	Activity A	–	3
B	Activity B	A	5
C	Activity C	A	2
D	Activity D	B, C	4
E	Activity E	D	3

1. Draw a fully labelled network diagram (node numbers, EST, LFT, letters and durations).
2. State the critical path and how long the project takes.
3. Which activity is non-critical? Calculate its float.

Draw your network diagram here (use pencil)

Question 2: Sweet Treats Bakery

Thandi is opening a new bakery in Claremont. She has listed the activities needed before opening day.

Activity	Description	Prerequisite(s)	Duration (days)
A	Find and rent premises	–	6
B	Hire staff	–	4
C	Buy ovens	A	3
D	Decorate the shop	A	4
E	Train staff	B	2
F	Stock up on ingredients	C	2
G	Opening day	D, E, F	1

1. Draw a fully labelled network diagram.
2. State the critical path and the minimum time to open the bakery.
3. Complete a float table for all activities (Duration, EST, LFT, Total float).
4. Thandi wants to open within 14 days. Can she? Explain.

Draw your network diagram here (use pencil)

Question 3: School Fun Day

The Grade 9 committee is organising a fun day to raise money for charity.

Activity	Description	Prerequisite(s)	Duration (weeks)
A	Plan the event	–	2
B	Book the venue	A	3
C	Find sponsors	A	4
D	Advertise the event	B	3
E	Order food	B	2
F	Sell tickets	C, D	4
G	Set up on the day	E, F	1

1. Draw a fully labelled network diagram.
2. State the critical path and how long the project will take.
3. Calculate the float for activities C and E.
4. Finding sponsors (C) now takes 3 weeks longer. Recalculate. Has the critical path changed? How long will the project take now?

Draw your network diagram here (use pencil)

Question 4: SnapStudy App Launch

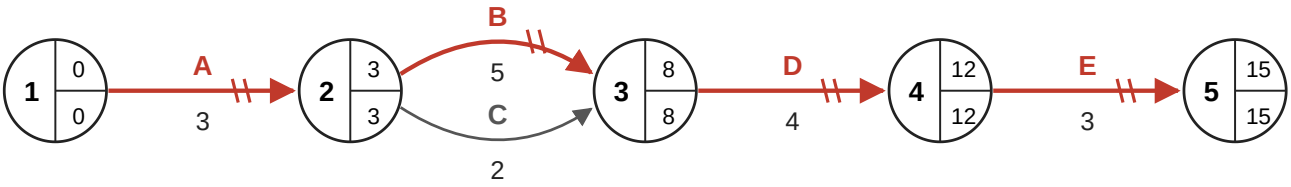
A small tech business is launching a study app for high-school learners. The owner must launch within 20 weeks or lose an investor.

Activity	Description	Prerequisite(s)	Duration (weeks)
A	Market research	–	4
B	Design the app	–	6
C	Hire developers	–	3
D	Build the app	B, C	8
E	Plan marketing	A	3
F	Test the app	D	4
G	Run marketing campaign	E	2
H	Launch	F, G	1

1. Draw a fully labelled network diagram.
2. State the critical path and the project length.
3. Will the app launch within 20 weeks? Show how you know.
4. Which activity has the most float? How could the owner use this float to help the project?

Draw your network diagram here (use pencil)

Question 1: Warm-up

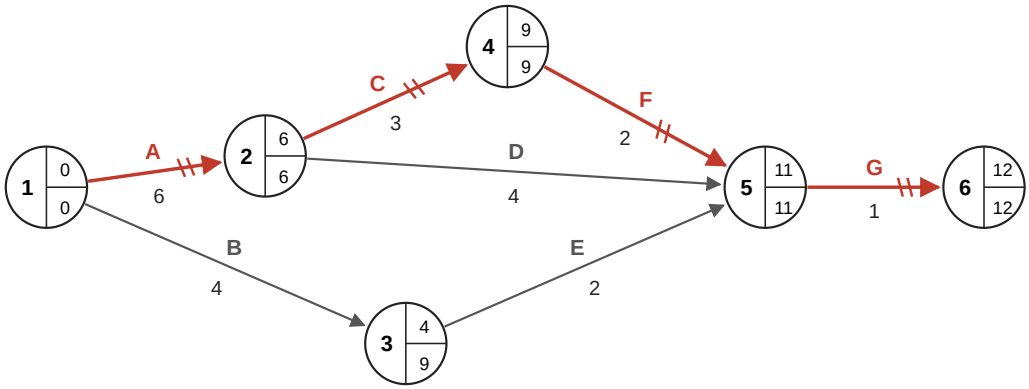


Critical path: A, B, D, E = **15 days**

Activity	Duration	EST	LFT	Float
A	3	0	3	0
B	5	3	8	0
C	2	3	8	3
D	4	8	12	0
E	3	12	15	0

C is non-critical: float = 8 − 3 − 2 = **3 days**.

Question 2: Sweet Treats Bakery

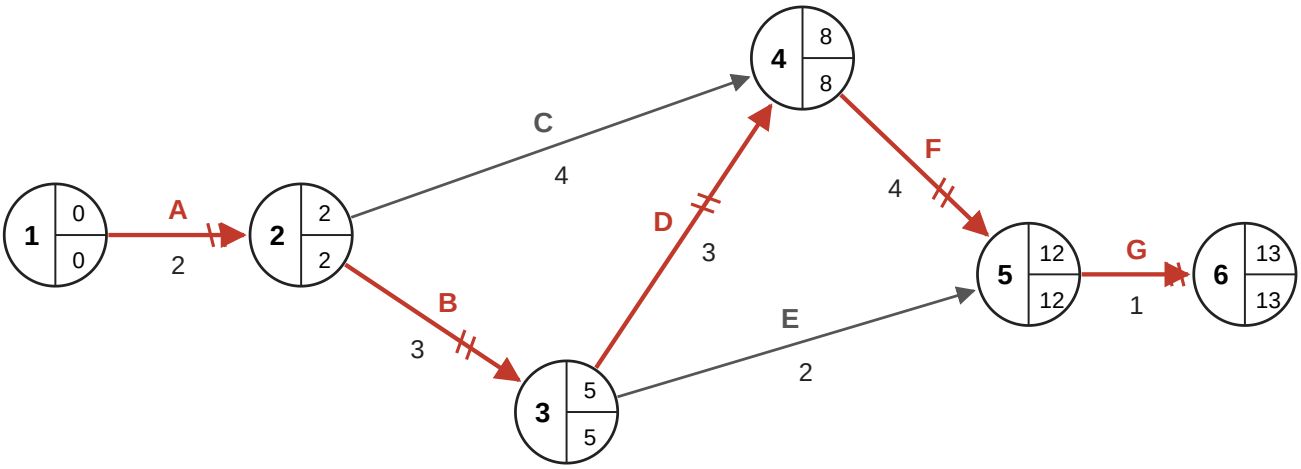


Critical path: A, C, F, G = **12 days**

Activity	Duration	EST	LFT	Float
A	6	0	6	0
B	4	0	9	5
C	3	6	9	0
D	4	6	11	1
E	2	4	11	5
F	2	9	11	0
G	1	11	12	0

Yes. The bakery takes 12 days, which is less than 14, so it opens 2 day(s) early. Float: A = 0, B = 5, C = 0, D = 1, E = 5, F = 0, G = 0.

Question 3: School Fun Day

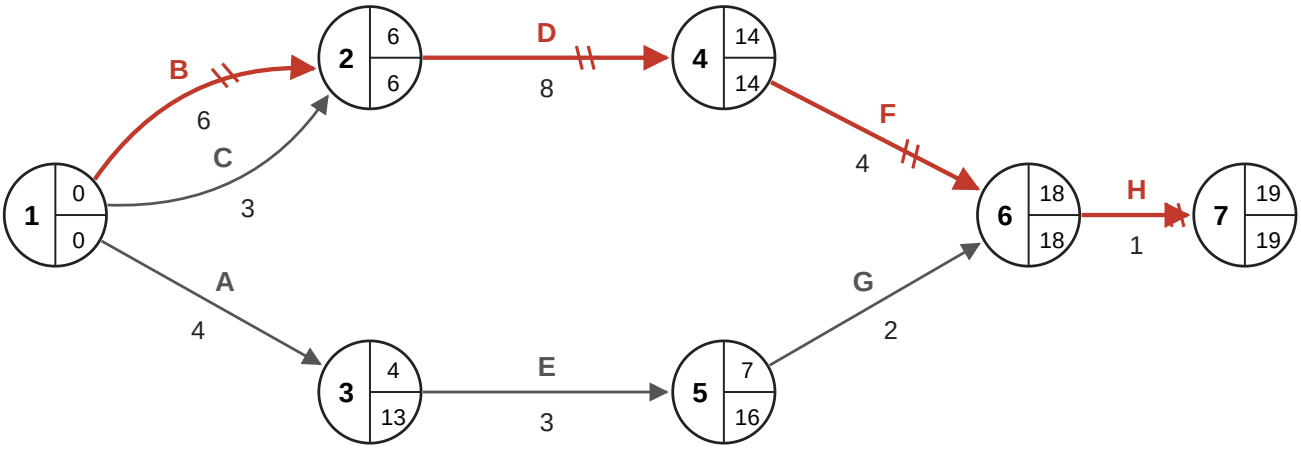


Critical path: A, B, D, F, G = **13 weeks**

Activity	Duration	EST	LFT	Float
A	2	0	2	0
B	3	2	5	0
C	4	2	8	2
D	3	5	8	0
E	2	5	12	5
F	4	8	12	0
G	1	12	13	0

Float: C = 2 weeks, E = 5 weeks.
What-if (C = 7 weeks): new critical path A, C, F, G = **14 weeks**. Yes, the critical path changed because C's extra 3 weeks is more than its 2 weeks of float.

Question 4: SnapStudy App Launch



Critical path: B, D, F, H = **19 weeks**

Activity	Duration	EST	LFT	Float
A	4	0	13	9
B	6	0	6	0
C	3	0	6	3
D	8	6	14	0
E	3	4	16	9
F	4	14	18	0
G	2	7	18	9
H	1	18	19	0

Yes. 19 weeks is less than 20, so it launches 1 week(s) early.
Most float: **A** (9 weeks; E and G share the same route). The owner could move workers from the marketing route onto building or testing the app, to protect the critical path.